Thin Film Interference



Review: Two ways to think about interference

1. Think in terms of **Phase Difference**

$$\Delta\phi = k\Delta x + \Delta\phi_0 = m(2\pi)$$ **Constructive Int.**

$$\Delta\phi = k\Delta x + \Delta\phi_0 = \left(m + \frac{1}{2}\right)(2\pi)$$ **Destructive Int.**

If $\Delta\phi_0 = 0$, meaning that the two sources are **Coherent,** then
Its possible to simplify the equation to

$$\Delta\phi = k\Delta x + \Delta\phi_0$$

$$\Delta\phi = \frac{2\pi\Delta x}{\lambda} + 0$$

$$\frac{2\pi}{\lambda}\Delta x = m(2\pi)$$

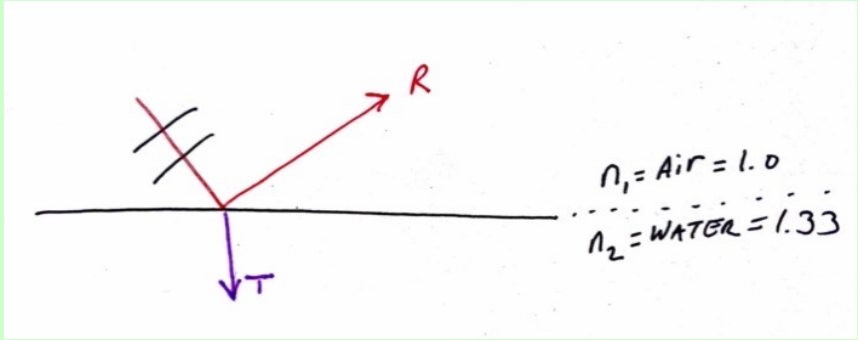
$$\frac{2\pi}{\lambda}\Delta x = \left(m + \frac{1}{2}\right)(2\pi)$$

$$\Delta x = m\lambda$$ **Const.**

$$\Delta x = \left(m + \frac{1}{2}\right)\lambda$$ **Dest.**

Thin Film Interference One application of 1D interference is when you have a thin film boundary between two materials

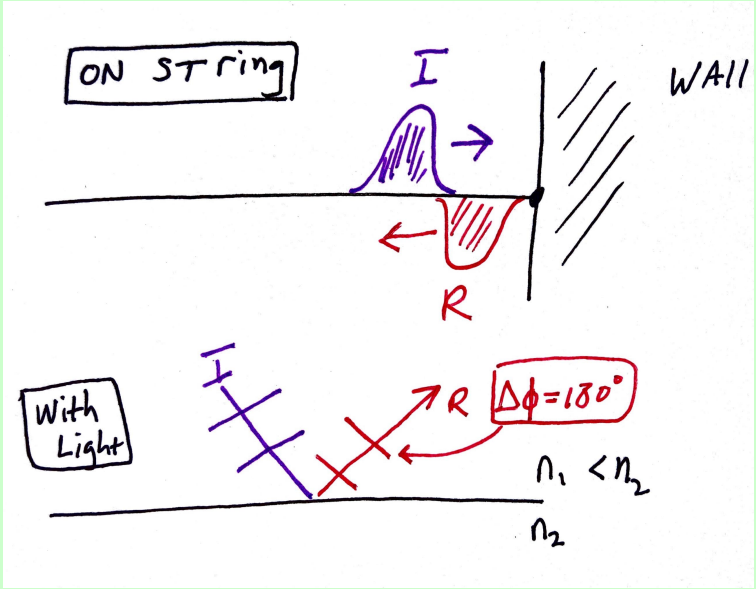
1. In the thin film material $\boxed{\lambda_n = \frac{\lambda_0}{n}}$



2. When waves hit discontinuities, it splits into reflected and transmitted modes.

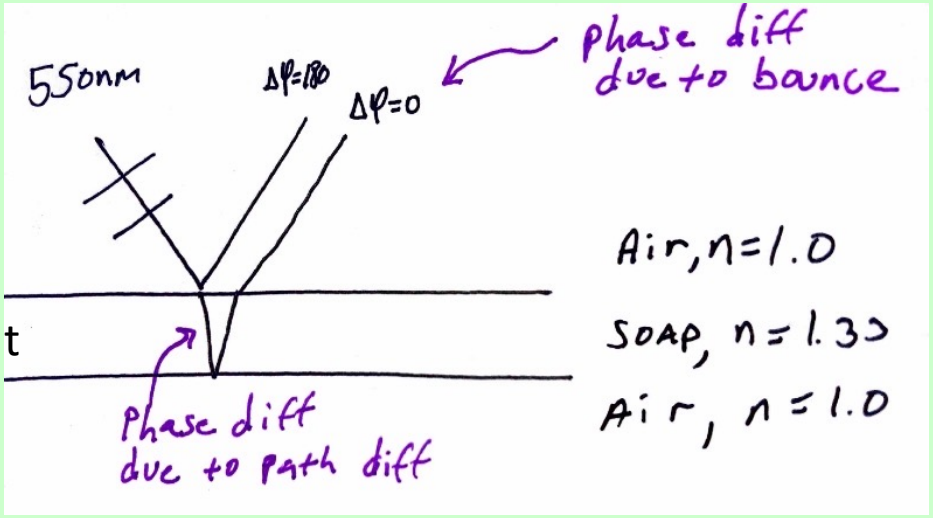
3. When wave bounces off **denser** matter, it gets a 180 degree phase shift

(if it bounces off less dense matter then no shift)



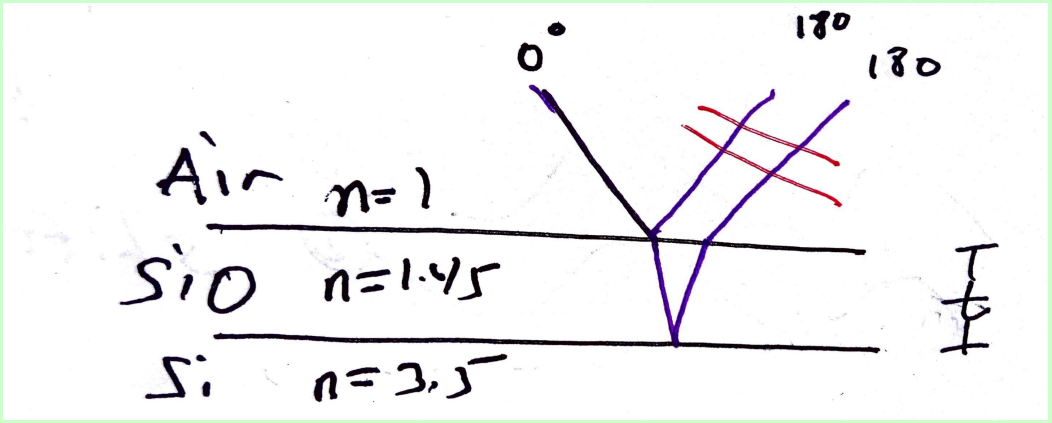
Ex. What happens at a double interface like a soap bubble?

What thickness to get constructive?

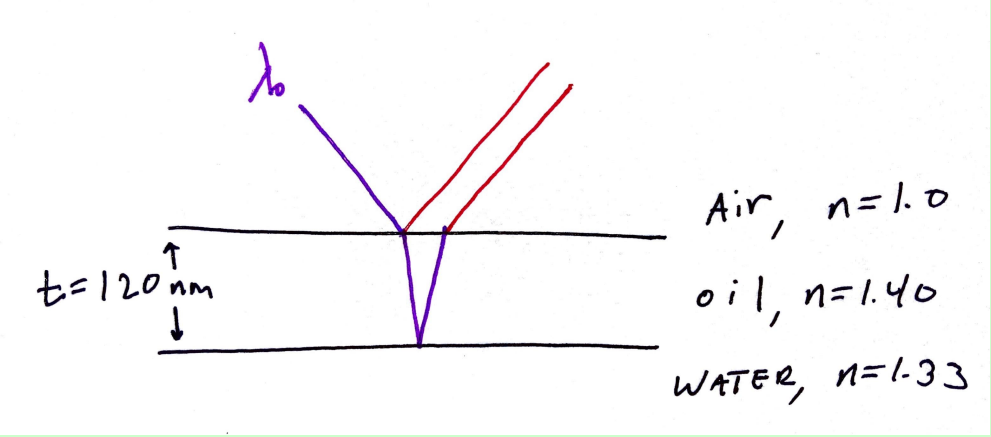


What thickness to get destructive?

Ex. Non-reflective coating is put on camera lenses. What thickness should you make it to cancel out 600 nm light?

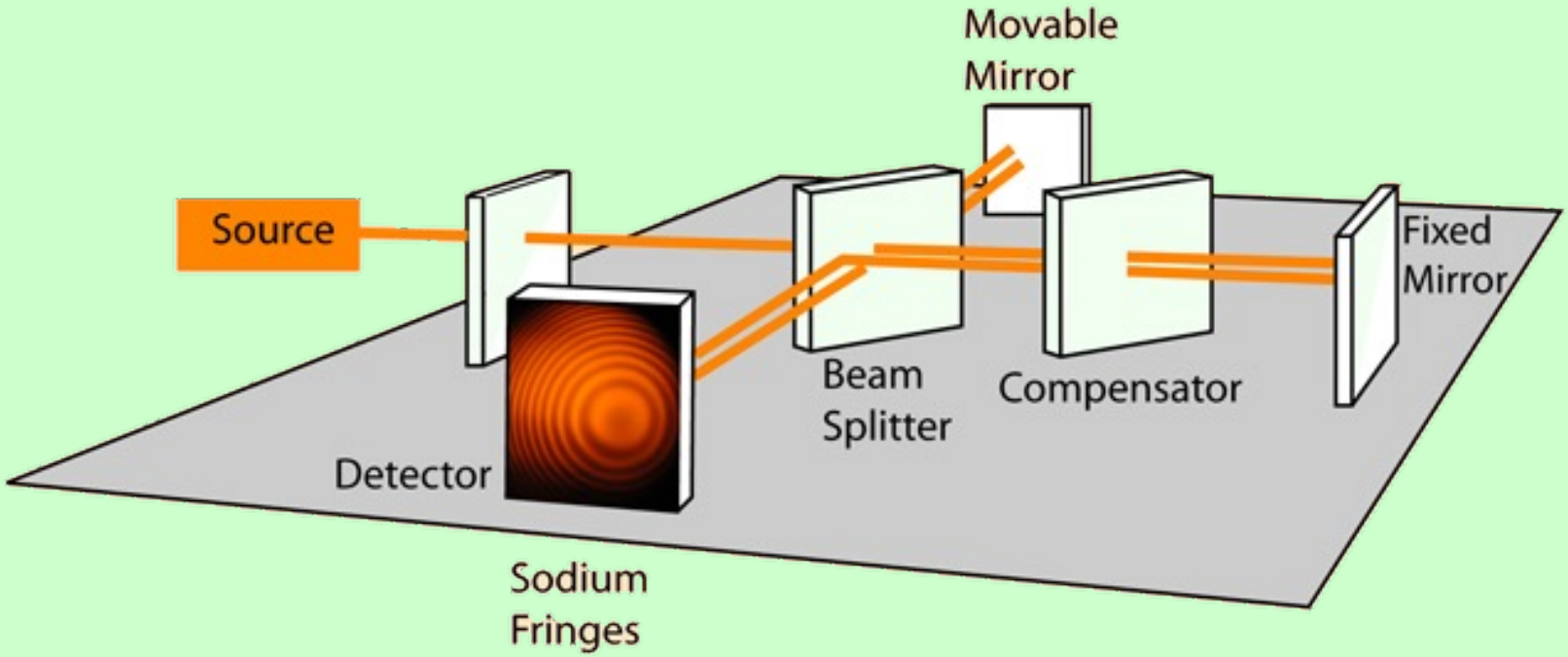


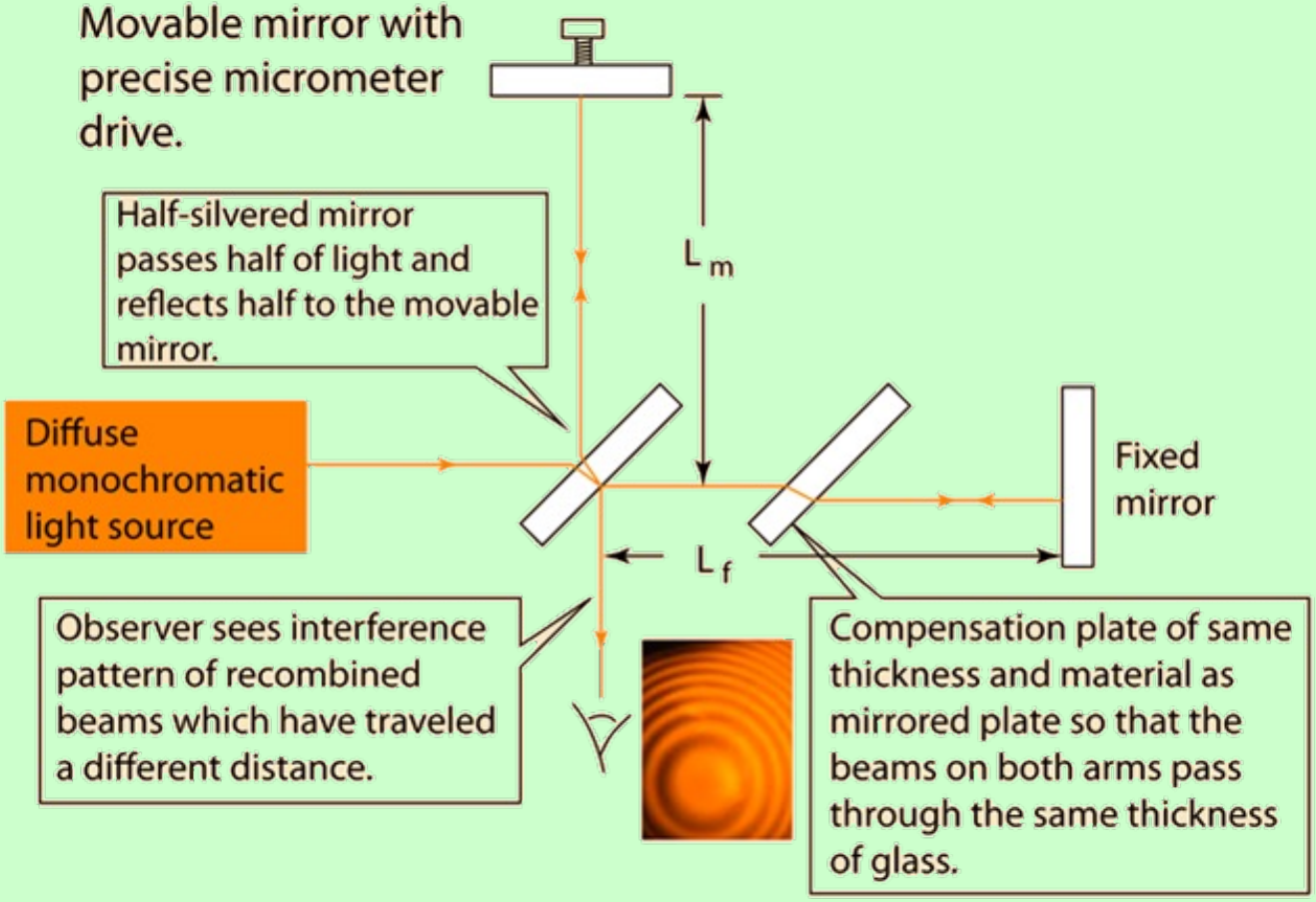
Ex. An oil slick is 120 nm thick, illuminated by white light. If oil has n=1.4, what color does the oil slick appear?



Michelson Interferometer:

Another application of 1D interference is the Michelson Interferometer. Which splits light Into two paths (one of which can be adjusted) and then interfers them





$$\Delta x = 2L_m - 2L_f = m\lambda$$ Constructive

So if you change the adjustable mirror path by $\Delta d = m\frac{\lambda_0}{2}$

The central fringe will move to replace the one m away



Ex. A Michelson Interferometer uses 630 nm laser source. How far must you move the adjustable mirror, so that a single fringe moves past a reference line?